U-slot array antenna report

For the first antenna topology (Antenna1), Fig. 1 shows a single element u-slot microstrip patch fed by a microstrip line that is designed and simulated. The patch dimension, slot dimension and the substrate thickness and permittivity is designed for optimum coupling.

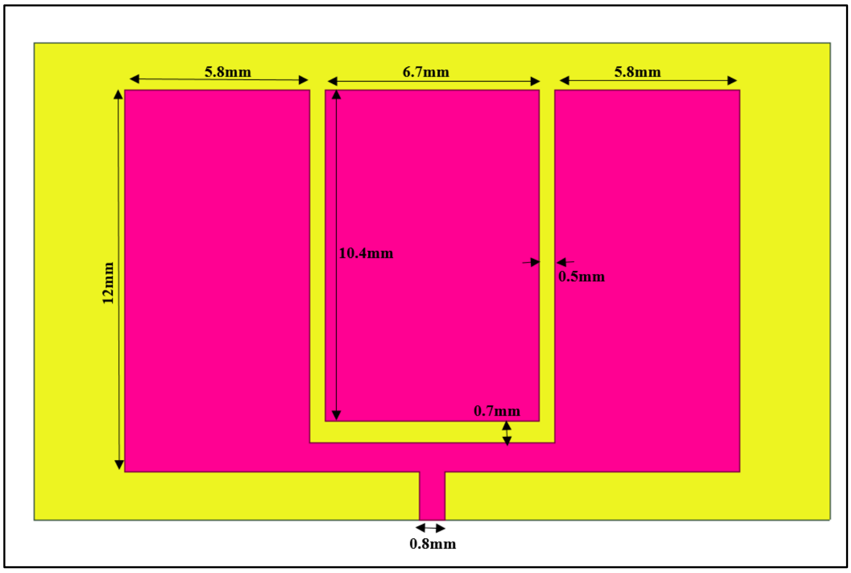


Fig.1. Single element u-slot microstrip patch antenna

The substrate, having a relative permittivity of 2.7 has dimensions of 25 mm in length, 15 mm in width, and 3.5 mm in thickness. The U-slot is incorporated into the patch with a slot width of 0.5 mm, while the horizontal arms of the U are 5.8 mm each. The central vertical arm has a length of 6.7 mm, and the feed line width is 0.8 mm. The patch extends 12 mm vertically on the substrate. These dimensions are critical for the design and optimization of the antenna's performance.

For the second antenna topology (Antenna2), Fig. 1 shows a single element u-slot microstrip patch fed by a microstrip line that is designed and simulated on a substrate with relative permittivity of 2.7 and thickness of 5mm

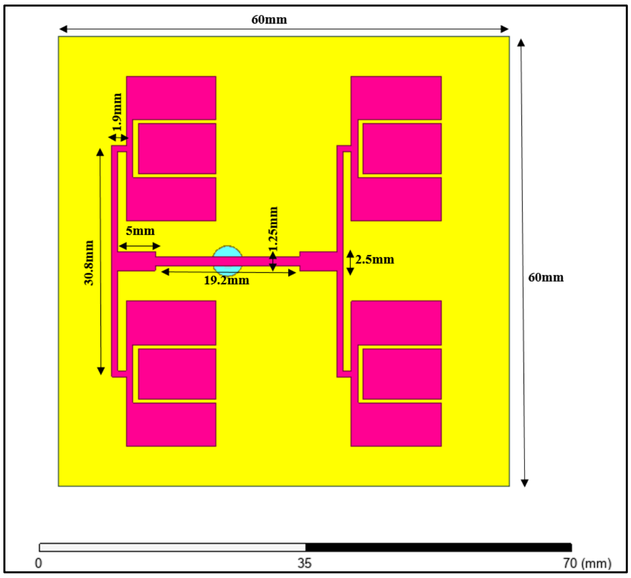


Fig.2. 2x2 array u-slot microstrip patch antenna

The figure depicts a U-slot microstrip patch array antenna that employs the same U-slot patch design as described previously. The array consists of four identical patches arranged symmetrically on a 60 mm × 60 mm substrate. Each patch has dimensions consistent with the earlier description, including a U-slot width of 0.5 mm, horizontal arm lengths of 5.8 mm, and a vertical arm length of 6.7 mm. The patches are spaced 30.8 mm apart vertically, while the feed network provides a separation of 19.2 mm horizontally between the rows. The feed line width is 2.5 mm, and the branch lines connecting to the patches have a width of 1.9 mm. This configuration is designed to optimize the array's performance and enhance its radiation characteristics.

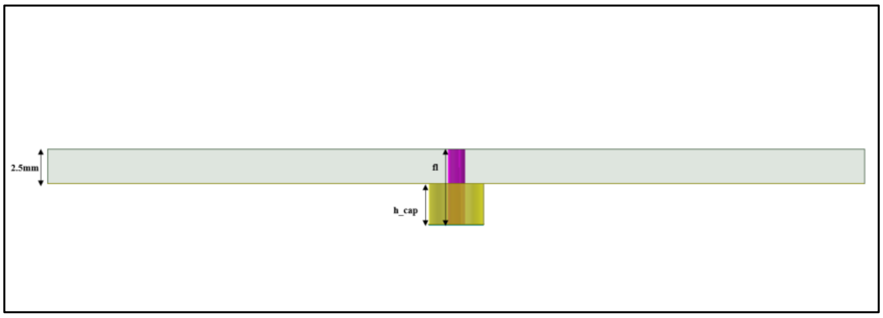


Fig.3. Side-view of the array antenna showing the feed line through the substrate.

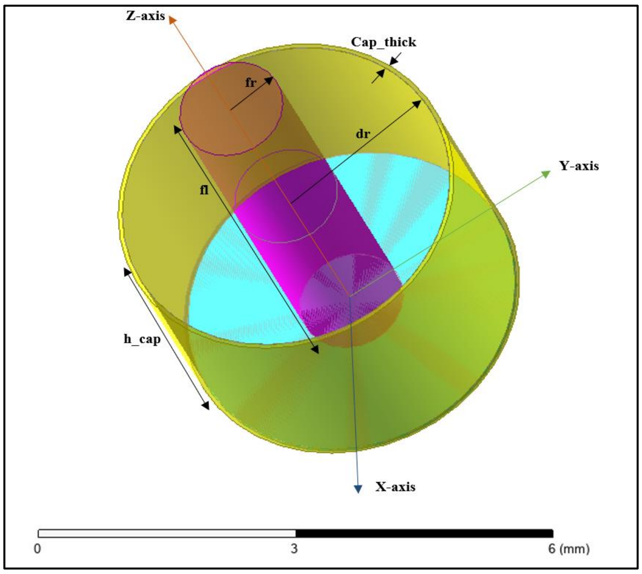


Fig.4. Trimetric view of the co-axial feed line.

The bottom-fed coaxial feed line for the U-slot microstrip patch array antenna consists of an inner conductor with a radius of 0.63 mm. This conductor is surrounded by a dielectric material made of Teflon, which has a radius of 2 mm. The coaxial feed line extends vertically from the ground plane, penetrating through the substrate to provide excitation to the patches. The Teflon dielectric ensures proper insulation and impedance matching, minimizing losses and enhancing the antenna's efficiency. This feed configuration is suitable for achieving uniform power distribution across the array and ensures robust performance at the desired frequency. The cap is terminated by a PEC layer with a thickness of 0.05mm.

The dimensions are tabulated in Table 1.

Table 1:Dimensions of the feed

Parameters	Description	Dimensions (mm)
fr	Radius of feedline	0.63
fl	Length of feedline	5.5
h-cap	Height of feed cap	3
dr	Dielectric radius	2
Cap-thick	Thickness of the cap	0.05

Results for Single element antenna:

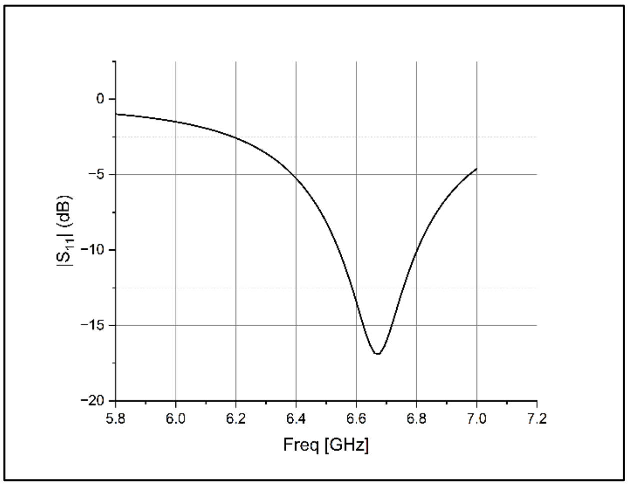
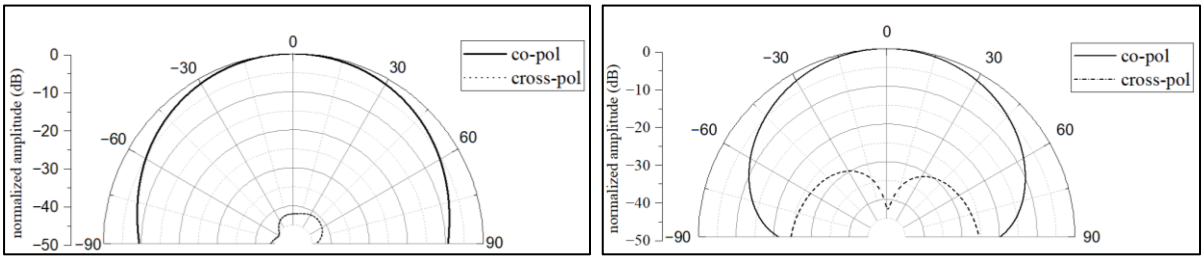


Fig. 5. Reflection coefficient vs frequency.

The resonant frequencies are obtained at 6.64 GHz with good coupling characteristics. Reflection coefficient -17 dB is achieved. The radiation patterns at the resonant frequency is shown in Fig. 6.



(a) $\phi = 0^{o}$ (b) $\phi = 90^{o}$

Fig. 6. Radiation pattern at 6.64 GHz (a) $\phi = 0^{o}$ (b) $\phi = 90^{o}$

The radiation pattern plotted in 3-dimensional spherical graph, gives a total gain of 6.45dB.

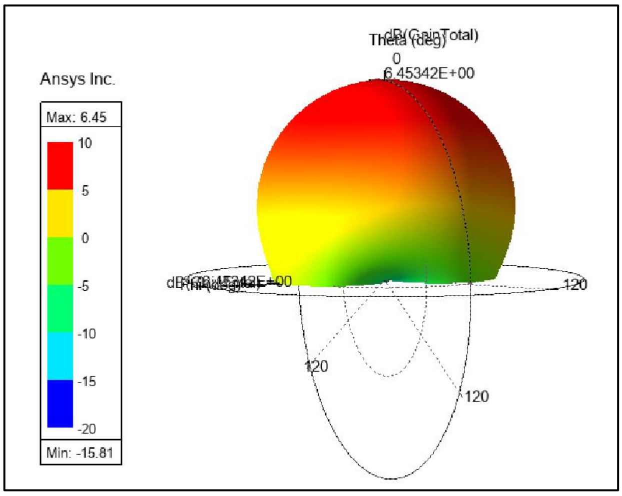


Fig. 7. Radiation pattern at 6.64 GHz

Results for array antenna:

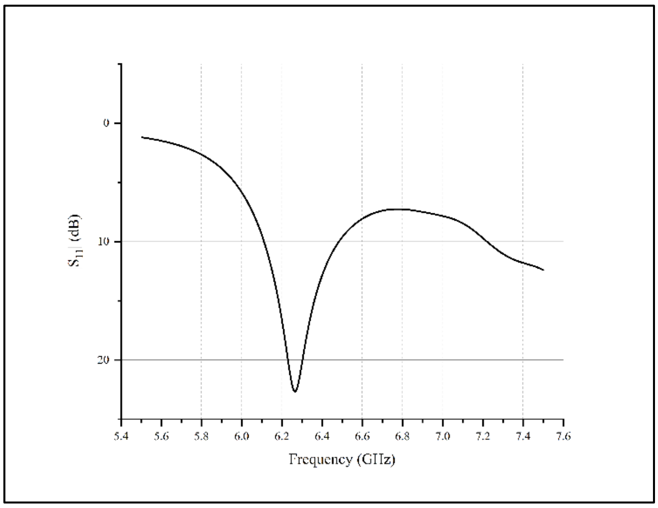
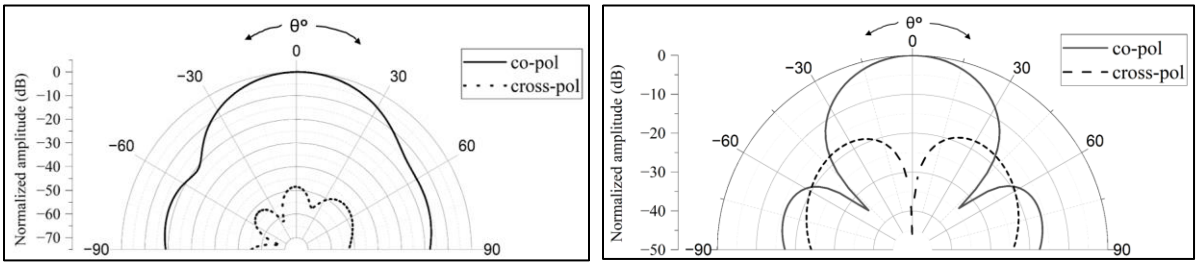


Fig. 8. Reflection coefficient vs frequency.

The resonant frequencies are obtained at 6.24 GHz with good coupling characteristics. Reflection coefficient -22.36 dB is achieved. The radiation patterns at the resonant frequency is shown in Fig. 6.



(a) $\phi = 0^{o}$ (b) $\phi = 90^{o}$

Fig. 9. Radiation pattern at 6.24 GHz (a) $\phi = 0^{o}$ *(b)* $\phi = 90^{o}$

The radiation pattern plotted in 3-dimensional spherical graph, gives a total gain of 12.57dB.

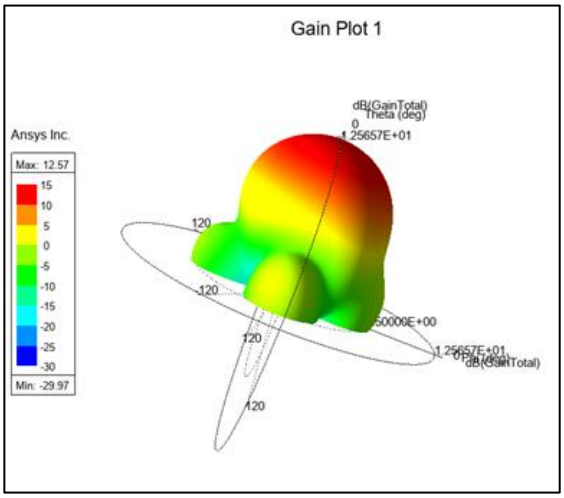


Fig. 10. Radiation pattern at 6.24 GHz